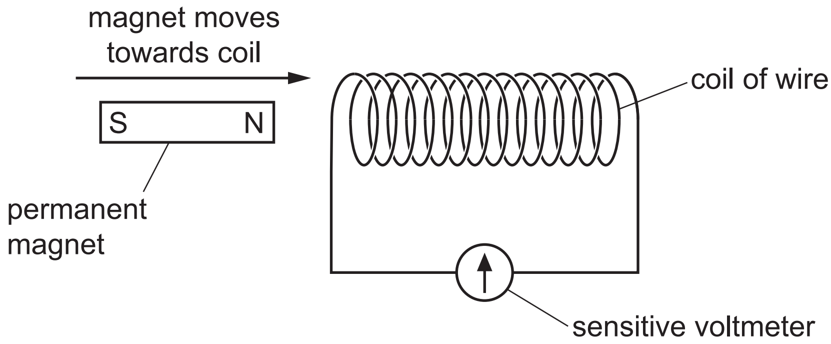
31 The diagram shows a magnet moving towards a coil of wire.



When the magnet moves towards the coil of wire, an electromotive force (e.m.f.) is induced.

Students are asked for one change that will increase the magnitude of the induced e.m.f.

Three changes are proposed.

1 Move the magnet more quickly.

2 Use a stronger magnet.

3 Increase the number of turns on the coil.

Which changes, on their own, will increase the magnitude of the induced e.m.f.?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

32 What is used to identify the pattern and direction of the magnetic field due to the current in a straight wire?

A an ammeter

B a compass

C iron filings

D a voltmeter

33 Which description of the structure of an atom is correct?

A negatively charged electrons surrounding an uncharged nucleus

B negatively charged electrons surrounding a positively charged nucleus

C positively charged electrons surrounding an uncharged nucleus

D positively charged electrons surrounding a negatively charged nucleus